

Software Requirements Specification (SRS)

AI Smart Surveillance System

Version: 1.0

1. Introduction

Purpose: Define requirements for an AI-powered surveillance platform for industrial training.

Scope: Live monitoring, AI detection, alerts, reports, user management.

2. Objectives

Real-time monitoring, AI-based detections, centralized administration, reporting, role-based access.

3. User Roles

Super Admin, Security Manager.

4. Functional Modules

Dashboard

- KPIs, charts, camera status, alerts.

Authentication & Users

- Login, RBAC, profile, audit log.

Camera Management

- Add/Edit/Delete cameras, RTSP/IP/USB, grouping, schedules.

Live Monitoring

- Camera view, Fullscreen, Snapshots, recording, overlays.

AI Detection

- **Person, Vehicle, Face (demo), Fire/Smoke, Abandoned Object.**

Alert Management

- Real-time alerts, acknowledge, assign, resolve, notifications.

Incident Management

- Create incidents, notes, attachments, lifecycle.

Recording & Media

- Recordings, snapshots, archive, download.

Reports & Analytics

- Daily/Weekly/Monthly, exports PDF/CSV/Excel, trends.

API

- JWT auth, REST endpoints, webhook support.

5. Technology Stack

Frontend: React/Next.js

Backend: Python FastAPI

AI: YOLOv8, OpenCV, MediaPipe

DB: PostgreSQL

Cache: Redis (optional)

Storage: Local

6. Database Entities

Users, Roles, Camera, Zones, AIModels, Alerts, Incidents, Recordings, Snapshots, Notifications, AuditLogs, Reports.

7. Acceptance Criteria

Users can view live feeds, receive AI alerts, review incidents, generate reports, and configure rules/settings.